

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OPERATING SYSTEMS COURSE CODE: CSA04

COURSE OUTCOME COVERED

CO2: Analyze process management and deadlock handling techniques to improve CPU

utilization and system performance(BL3-BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 10 Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I M.V.Priyavarsheni (192524428)_____ (Name / Reg No) certify that this submission is my original

work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student: M V.Priyavarsheni

1. An airport has five check-in counters serving passengers. Each passenger arrival is considered a process with varying arrival and service

times. During peak hours, passengers with special assistance are given priority.

Airport Check-in Counter Scheduling Using Priority Scheduling

An airport has five check-in counters serving passengers. In this system, each passenger is considered a process. Passengers arrive at different times and require different amounts of service time to complete the check-in procedure. During peak hours, the airport must manage long queues efficiently while ensuring that passengers who require special assistance receive timely service.

To achieve this, the airport uses the Priority Scheduling Algorithm. In this algorithm, every passenger is assigned a priority level. Passengers requiring special assistance, such as elderly people, passengers with disabilities, pregnant women, or those with medical emergencies, are given higher priority than regular passengers. The scheduler always selects the highest-priority passenger from the waiting queue for service.

Working of the System:

1. Passengers arrive at the airport and join the check-in queue.
2. Each passenger is assigned an arrival time, service time, and priority level.
3. The five check-in counters continuously check for waiting passengers.
4. The passenger with the highest priority is selected first for service.
5. If two passengers have the same priority, the one who arrived earlier is served first using the First-Come, First-Served (FCFS) principle.
6. After a passenger completes check-in, the counter becomes available for the next passenger.
7. This process continues until all passengers have been served.

Example:

Process	Arrival Time	Service Time	Priority
P1	0	5	2
P2	1	3	1
P3	2	4	3
P4	3	2	1
P5	4	6	2

Here, a lower priority number indicates higher priority.

Order of Service: P2 → P4 → P1 → P5 → P3

Advantages of Priority Scheduling at Airport Check-in Counters:

- Provides faster service to passengers who need special assistance.
- Reduces waiting time for high-priority passengers.
- Improves overall customer satisfaction.
- Utilizes check-in counters efficiently during busy hours.
- Helps airport staff manage passenger flow in an organized manner.

Disadvantages:

- Low-priority passengers may experience longer waiting times.
- Starvation may occur if high-priority passengers continue to arrive frequently.

Conclusion:

Priority Scheduling is an effective method for managing airport check-in counters, especially during peak hours. By giving preference to passengers requiring special assistance, the airport can provide better service, reduce delays, and improve operational efficiency while maintaining fairness among passengers with the same priority level.

2. Analyse the scheduling using FCFS and Priority Scheduling. Determine the waiting time and turnaround time for each passenger and identify which algorithm provides better service efficiency.

Analysis of FCFS and Priority Scheduling

Process	Arrival Time (AT)	Service Time (BT)	Priority
P1	0	5	2
P2	1	3	1
P3	2	4	3
P4	3	2	1
P5	4	6	2

Lower priority number indicates higher priority.

1. FCFS (First Come First Served)

Gantt Chart

0 — P1 — 5 — P2 — 8 — P3 — 12 — P4 — 14 — P5 — 20

Calculations

Process	AT	BT	Completion Time (CT)	Waiting Time (WT)	Turnaround Time (TAT)
P1	0	5	5	0	5
P2	1	3	8	4	7
P3	2	4	12	6	10
P4	3	2	14	9	11
P5	4	6	20	10	16

Average Waiting Time = $(0 + 4 + 6 + 9 + 10) / 5 = 5.8$ units

Average Turnaround Time = $(5 + 7 + 10 + 11 + 16) / 5 = 9.8$ units

2. Priority Scheduling (Non Pre -emptive)

Priority Order: P1 → P2 → P4 → P5 → P3

Gantt Chart

0 — P1 — 5 — P2 — 8 — P4 — 10 — P5 — 16 — P3 — 20

Calculations

Process	AT	BT	Completion Time (CT)	Waiting Time (WT)	Turnaround Time (TAT)
P1	0	5	5	0	5
P2	1	3	8	4	7
P4	3	2	10	5	7
P5	4	6	16	6	12
P3	2	4	20	14	18

Average Waiting Time = $(0 + 4 + 5 + 6 + 14) / 5 = 5.8$ units

Average Turnaround Time = $(5 + 7 + 7 + 12 + 18) / 5 = 9.8$ units

Comparison

- FCFS serves passengers according to their arrival time without considering priority. It is simple and fair but may delay passengers who require urgent assistance.
- Priority Scheduling gives preference to passengers with higher priority, such as elderly or disabled passengers. This improves service quality for special-assistance passengers and ensures they are served earlier.
- In this example, both FCFS and Priority Scheduling have the same average waiting time (5.8 units) and average turnaround time (9.8 units).

Conclusion

However, Priority Scheduling is more suitable for airport check-in counters because it provides better service to passengers who need special assistance. Therefore, Priority Scheduling is the preferred algorithm for improving service efficiency in this airport scenario.

A hospital emergency room receives patients continuously. Critical patients must be treated immediately, while non-critical patients wait in a queue.

Model this as a process scheduling problem using Priority Scheduling and Round Robin (time quantum = 5 minutes). Compare the response time for critical and non-critical patients.

Hospital Emergency Room Scheduling Using Priority Scheduling and Round Robin

A hospital emergency room continuously receives patients. Each patient is considered a process. Critical patients require immediate medical attention, while non-critical patients can wait. This scenario can be modelled using Priority Scheduling and Round Robin Scheduling (Time Quantum = 5 minutes).

Patient Details

Patient	Arrival Time (min)	Treatment Time (min)	Priority
P1 (Critical)	0	10	1
P2 (non-critical)	1	8	2
P3 (Critical)	2	6	1

P4 (non-critical)	3	12	2
-------------------	---	----	---

Lower priority number indicates higher priority.

1. Priority Scheduling

Critical patients are treated before non-critical patients.

Execution Order

P1 → P3 → P2 → P4

Response Time Calculation

Patient	Arrival Time	Start Time	Response Time
P1 (Critical)	0	0	0 min
P3 (Critical)	2	10	8 min
P2 (non-critical)	1	16	15 min
P4 (non-critical)	3	24	21 min

Average Response Time

Patient Type	Average Response Time
Critical Patients	$(0 + 8) / 2 = 4 \text{ min}$
Non-Critical Patients	$(15 + 21) / 2 = 18 \text{ min}$

2. Round Robin Scheduling (Time Quantum = 5 Minutes)

Execution Sequence

P1 → P2 → P3 → P4 → P1 → P2 → P3 → P4 → P4

Response Time Calculation

Patient	Arrival Time	First Treatment Time	Response Time
---------	--------------	----------------------	---------------

P1 (Critical)	0	0	0 min
P2 (Non-Critical)	1	5	4 min
P3 (Critical)	2	10	8 min
P4 (Non-Critical)	3	15	12 min

Average Response Time

Patient Type	Average Response Time
Critical Patients	$(0 + 8) / 2 = 4 \text{ min}$
Non-Critical Patients	$(4 + 12) / 2 = 8 \text{ min}$

Comparison Table

Criteria	Priority Scheduling	Round Robin (Q = 5 min)
Critical Patient Response Time	4 min	4 min
Non-Critical Patient Response Time	18 min	8 min
Fairness	Low	High
Starvation Possibility	Yes	No
Service for Critical Patients	Excellent	Good
Service for Non-Critical Patients	Poor	Better
Best Use Case	Emergency treatment	Balanced patient handling

Conclusion

Priority Scheduling is best when critical patients must receive immediate attention because it minimizes their waiting and response times. However, non-critical patients may experience long delays. Round Robin Scheduling provides fairer service by allowing every patient to receive treatment in turns, reducing the response time for non-critical patients. Therefore, Priority Scheduling is preferred for emergency rooms where critical cases must be treated first, while Round Robin is suitable when fairness among all patients is important.

4. A cloud service provider receives ten computational tasks from clients. Each task requires different CPU burst times and arrives at different intervals. Apply FCFS, SJF, and Round Robin scheduling to determine the

average waiting time and CPU utilization. Recommend the best scheduling algorithm for the cloud environment.

Answer

A cloud service provider receives computational tasks from clients. Each task is treated as a process with its own CPU burst time and arrival time. The objective is to minimize waiting time and maximize CPU utilization.

Assumed Task Details

Task	Burst Time (ms)
T1	10
T2	5
T3	8
T4	6
T5	2
T6	4
T7	7
T8	3
T9	9
T10	5

FCFS (First Come First Served)

Task	Waiting Time
T1	0
T2	10
T3	15
T4	23
T5	29
T6	31
T7	35
T8	42
T9	45
T10	54

Average Waiting Time = 28.4 ms

CPU Utilization $\approx$ 100%

SJF (Shortest Job First)

Execution Order:

T5 → T8 → T6 → T2 → T10 → T4 → T7 → T3 → T9 → T1

Average Waiting Time = 19.5 ms

CPU Utilization ≈ 100%

Round Robin (Time Quantum = 4 ms)

Average Waiting Time ≈ 24.8 ms

CPU Utilization ≈ 95–98%

Comparison Table

Algorithm	Average Waiting Time	CPU Utilization
FCFS	28.4 ms	100%
SJF	19.5 ms	100%
Round Robin	24.8 ms	95–98%

Conclusion

SJF provides the lowest average waiting time and high CPU utilization. Therefore, **Shortest Job First (SJF)** is the best scheduling algorithm for a cloud environment.

5. A smart city uses sensors to detect vehicle density at an intersection. Each traffic lane is represented as a process competing for signal access.

Design a scheduling strategy that minimizes vehicle waiting time. Compare the effectiveness of Round Robin and Priority Scheduling for emergency vehicles.

Answer

In a smart city traffic system, each lane is considered a process competing for access to the traffic signal. Sensors continuously monitor vehicle density and traffic conditions.

Scheduling Strategy

1. Use sensors to measure vehicle density in each lane.
2. Assign higher priority to lanes with heavy traffic.
3. Give the highest priority to emergency vehicles such as ambulances, fire trucks, and police vehicles.
4. Dynamically adjust green signal duration based on traffic conditions.
5. Continuously update priorities to reduce congestion and waiting time.

Round Robin Scheduling

- Each lane receives a fixed green signal time.
- Lanes are served in cyclic order.
- Ensures fairness among all lanes.

Priority Scheduling

- Lanes with higher traffic density receive priority.
- Emergency vehicles are given immediate access.
- Signal timing changes dynamically based on traffic conditions.

Comparison Table

Criteria	Round Robin	Priority Scheduling
Fairness	High	Medium
Vehicle Waiting Time	Moderate	Low
Emergency Vehicle Handling	Poor	Excellent
Congestion Control	Average	Excellent
Starvation Risk	No	Possible
Implementation Complexity	Low	High
Traffic Flow Efficiency	Moderate	High

Conclusion

Priority Scheduling is more effective for smart city traffic management because it minimizes vehicle waiting time, reduces congestion, and provides immediate access to emergency vehicles. Although Round Robin ensures fairness, Priority Scheduling offers better overall traffic efficiency.

6. During an online examination, thousands of students submit answers simultaneously. The server processes requests based on arrival times and submission priorities.

Construct a process scheduling table for eight requests and calculate Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) using FCFS and SJF Scheduling.

Answer

Request Details

Request	Arrival Time (AT)	Burst Time (BT)
R1	0	6
R2	1	4
R3	2	8
R4	3	3
R5	4	5
R6	5	2
R7	6	7
R8	7	4

FCFS Scheduling

Gantt Chart

0 | R1 | 6 | R2 | 10 | R3 | 18 | R4 | 21 | R5 | 26 | R6 | 28 | R7 | 35 | R8 | 39

Calculations

Request	AT	BT	CT	TAT = CT - AT	WT = TAT - BT
R1	0	6	6	6	0
R2	1	4	10	9	5
R3	2	8	18	16	8
R4	3	3	21	18	15
R5	4	5	26	22	17
R6	5	2	28	23	21

R7	6	7	35	29	22
R8	7	4	39	32	28

Average Waiting Time = 14.5 units

Average Turnaround Time = 19.4 units

SJF Scheduling

Execution Order

R1 → R6 → R4 → R2 → R8 → R5 → R7 → R3

Calculations

Request	AT	BT	CT	TAT	WT
R1	0	6	6	6	0
R6	5	2	8	3	1
R4	3	3	11	8	5
R2	1	4	15	14	10
R8	7	4	19	12	8
R5	4	5	24	20	15
R7	6	7	31	25	18
R3	2	8	39	37	29

Average Waiting Time = 10.75 units

Average Turnaround Time = 15.6 units

Conclusion

SJF provides lower average waiting time and turnaround time than FCFS, making it more efficient for processing online examination submissions.

7. A factory assembly line consists of multiple robotic arms, each performing a task. If one robot fails, the operating system must reschedule pending tasks. Analyse how process states (Ready, Running, Waiting, Terminated) change during task execution and explain how scheduling ensures continuous production.

Answer

In a factory assembly line, each robotic arm performs a specific task such as welding, painting, assembling, or packaging. These tasks can be modelled as processes managed by the operating system.

Process States

State	Description
Ready	Task is waiting for CPU allocation.
Running	Task is currently being executed.
Waiting	Task is waiting for resources or robot recovery.
Terminated	Task has completed execution.

State Transition

Ready → Running → Waiting → Running → Terminated

Example

1. Robot A receives a welding task and enters the Ready state.
2. The scheduler allocates CPU time, moving the task to Running.
3. If Robot A fails, the task enters the Waiting state.
4. The scheduler assigns the pending task to another available robot.
5. The task resumes Running and eventually reaches the Terminated state.

Role of Scheduling

- Detects robot failures.
- Reassigns pending tasks to available robots.
- Prevents production delays.
- Balances workload among robotic arms.
- Maintains continuous production flow.

Conclusion

Efficient scheduling enables uninterrupted manufacturing by reallocating tasks whenever a robot fails, ensuring maximum productivity and resource utilisation

8. A video streaming service processes user requests for video buffering. Premium users are assigned higher priority than standard users.

Given a set of user requests with arrival and burst times, implement Priority Scheduling and evaluate fairness, starvation, and throughput.

Answer

User Request Details

User	Arrival Time	Burst Time	Priority
U1 (Premium)	0	5	1
U2 (Standard)	1	4	3
U3 (Premium)	2	3	1
U4 (Standard)	3	6	3
U5 (Premium)	4	2	1

Lower priority number indicates higher priority.

Priority Scheduling Order

U1 → U3 → U5 → U2 → U4

Scheduling Table

User	AT	BT	CT	TAT	WT
U1	0	5	5	5	0
U3	2	3	8	6	3
U5	4	2	10	6	4
U2	1	4	14	13	9
U4	3	6	20	17	11

Evaluation

Fairness

- Premium users receive faster service.
- Standard users experience longer waiting times.
- Fairness is reduced because higher-priority users are favored.

Starvation

- Continuous arrival of premium users may indefinitely delay standard users.
- Therefore, starvation is possible.

Throughput

Throughput = Number of Requests Completed / Total Time

= 5 / 20

= **0.25 requests per unit time**

Comparison Table

Parameter	Observation
Fairness	Moderate
Starvation Risk	High
Throughput	Good
Premium User Experience	Excellent
Standard User Experience	Moderate to Poor

Conclusion

Priority Scheduling improves response time for premium users and increases customer satisfaction for high-priority clients. However, it may cause starvation of standard users. Techniques such as aging can be used to improve fairness while maintaining good throughput.

9.A bank processes ATM withdrawals, deposits, and fund transfers concurrently. Each transaction is treated as a process with varying execution times. Using Round Robin Scheduling (time quantum = 4 ms), determine the execution order and calculate average turnaround time and waiting time for ten transactions.

Answer

Transaction Details

Transaction	Burst Time (ms)
T1	10
T2	5
T3	8
T4	6
T5	12
T6	4
T7	7
T8	9
T9	3
T10	5

Execution Order

T1 → T2 → T3 → T4 → T5 → T6 → T7 → T8 → T9 → T10 → T1 → T2 → T3 → T4 → T5 → T7 → T8 → T10 → T1
→ T5 → T8

Results

Transaction	Waiting Time (ms)	Turnaround Time (ms)
T1	53	63
T2	40	45
T3	45	53
T4	46	52
T5	57	69
T6	20	24
T7	51	58
T8	57	66
T9	32	35
T10	48	53

Average Values

Average Waiting Time = 44.9 ms

Average Turnaround Time = 51.8 ms

10. An autonomous vehicle executes multiple processes simultaneously: obstacle detection, GPS navigation, speed control, and lane detection.

Identify the process states of each task at different time intervals and justify the use of pre-emptive scheduling in ensuring passenger safety.

Answer

Process State Analysis

Time Interval	Obstacle Detection	GPS Navigation	Speed Control	Lane Detection
t0	Running	Ready	Ready	Ready
t1	Running	Waiting	Ready	Ready
t2	Running	Ready	Running	Waiting
t3	Running	Ready	Ready	Running
t4	Terminated	Running	Ready	Ready

State Explanation

- **Ready:** Process is waiting for CPU allocation.
- **Running:** Process is currently executing.
- **Waiting:** Process is waiting for sensor data or resources.
- **Terminated:** Process execution has completed.

Why Pre-emptive Scheduling?

1. Obstacle detection must immediately interrupt less important tasks.
2. Emergency braking requires instant CPU access.
3. GPS updates can be delayed without affecting safety.
4. Prevents dangerous delays in critical operations.

Conclusion

Pre-emptive scheduling ensures that safety-critical processes always receive immediate CPU attention, improving passenger safety and system reliability.

11. A satellite control centre monitors communication, navigation, power management, and telemetry systems. Communication tasks have the

highest priority during emergencies. Simulate process execution using Priority Scheduling and analyse the impact of process starvation. Propose a solution using aging techniques to improve fairness.

Answer

Process Details

Process	Burst Time	Priority
Communication	8	1
Navigation	6	2
Power Management	5	3
Telemetry	4	4

Lower number indicates higher priority.

Execution Order

Communication → Navigation → Power Management → Telemetry

Priority Scheduling Table

Process	Burst Time	Priority	Completion Time
Communication	8	1	8
Navigation	6	2	14
Power Management	5	3	19
Telemetry	4	4	23

Impact of Starvation

- Low-priority tasks may wait indefinitely.
- Telemetry updates can be delayed during continuous emergencies.
- System fairness decreases.

Aging Technique

Aging gradually increases the priority of waiting processes.

Example:

Waiting Time	New Priority
0 min	4
5 min	3
10 min	2
15 min	1

Conclusion

Aging prevents starvation by ensuring long-waiting processes eventually receive CPU time while still prioritizing emergency communication tasks.

11. SJF (Non-Pre-emptive)

Process	Burst Time (ms)
P1	10
P2	4
P3	6
P4	2

a) Calculate Waiting Time and Turnaround Time

Execution Order

P4 → P2 → P3 → P1

Gantt Chart

0 – P4 – 2 – P2 – 6 – P3 – 12 – P1 – 22

Process	BT	WT	TAT
P4	2	0	2
P2	4	2	6
P3	6	6	12
P1	10	12	22

b) Average Waiting Time

$$= (0 + 2 + 6 + 12) / 4$$

$$= 5 \text{ ms}$$

Average Turnaround Time

$$= (2 + 6 + 12 + 22) / 4$$

$$= 10.5 \text{ ms}$$

c) Interpretation

SJF reduces average waiting time and turnaround time compared to FCFS, thereby improving CPU efficiency and overall system performance.

13. Round Robin Scheduling (Time Quantum = 4 ms)

Process	Burst Time (ms)
P1	20
P2	5
P3	13
P4	8

a) Gantt Chart

0–4 P1 | 4–8 P2 | 8–12 P3 | 12–16 P4 | 16–20 P1 | 20–21 P2 | 21–25 P3 | 25–29 P4 | 29–33 P1 | 33–37 P3 | 37–41 P1 | 41–42 P3 | 42–46 P1

b) Waiting Time and Turnaround Time

Process	BT	CT	TAT	WT
P1	20	46	46	26
P2	5	21	21	16
P3	13	42	42	29
P4	8	29	29	21

Average Waiting Time

$$= (26 + 16 + 29 + 21) / 4$$

$$= 23 \text{ ms}$$

Average Turnaround Time

$$= (46 + 21 + 42 + 29) / 4$$

$$= 34.5 \text{ ms}$$

c) Effect of Increasing Quantum to 10 ms

Parameter	Quantum = 4 ms	Quantum = 10 ms
Context Switching	High	Low
CPU Utilization	Lower	Higher
Response Time	Better	Slightly Worse
Scheduling Overhead	High	Low

Conclusion: A larger quantum reduces context switching and improves CPU utilization but decreases responsiveness.

14. CPU Utilization – Multiprogramming Analysis

- Probability of waiting for I/O (p) = 0.8
- CPU usage probability = 0.2
- Formula: $U = 1 - p^n$

a) Degree of Multiprogramming = 4

$$U = 1 - (0.8)^4$$

$$U = 1 - 0.4096$$

$$U = 0.5904$$

$$\text{CPU Utilization} = 59.04\%$$

b) Degree of Multiprogramming = 6

$$U = 1 - (0.8)^6$$

$$U = 1 - 0.262144$$

$$U = 0.737856$$

CPU Utilization = 73.79%

c) Interpretation

Degree of Multiprogramming	CPU Utilization
4 Processes	59.04%
6 Processes	73.79%

Increasing the degree of multiprogramming from 4 to 6 improves CPU utilization because when some processes wait for I/O, other processes can use the CPU. Therefore, the configuration with 6 processes provides better system performance and resource utilization.

